Cryptographic Proof and Integrity Report: Anonymo AI Framework

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1. Introduction

This report documents the complete cryptographic chain of custody implemented within the Anonymo AI Framework. The methodology establishes a verifiable, tamper-evident record of digital provenance for hybrid assets (created through equal collaboration between human and AI).

By calculating SHA-256 hashes of the raw visual binaries *before* injecting custom XMP metadata, the framework ensures that the cryptographic proof represents the pure visual content. This document serves as a public ledger of all hashes associated with the project's artifacts.

2. Methodology: The 4-Step Integrity Workflow

The integrity of the assets is guaranteed through a strict operational sequence:

1. **Generation:** The visual asset is generated via the hybrid workflow (Chris Montgomery & Manus AI).
2. **Pre-Injection Hashing:** A SHA-256 hash is calculated on the raw `.png` binary. This is the true "fingerprint" of the image.
3. **XMP Injection:** The custom `anonymo:` namespace metadata—including the pre-calculated hash, ORCID, and DOI—is injected into the image, creating a new `.jpg` file.
4. **Publication:** The final assets are published to GitHub (timestamp) and Zenodo (immutable DOI).

3. Cryptographic Ledger

The following tables record the exact SHA-256 hashes for all critical artifacts in the repository.

3.1. Visual Assets (Pre-Injection vs. Post-Injection)

This table demonstrates the core of the methodology. The "Pre-Injection Hash" is the value embedded inside the XMP metadata of the final file. If the visual content is ever questioned, one can strip the metadata from the final `.jpg` and recalculate the hash to match the pre-injection value.

Asset Name	Pre-Injection Hash (Raw PNG)	Post-Injection Hash (F
`anonymo_framework_24x24`	`afc12de617fbcc553801eefa99b60936e8907392689d3f41e7a17c88bf8d4b50`	`d60deaf817cb837194f`
`slide_img_spectrum`	`09f52736392185b48efb7667b2c4e4a0654504682ae39492c6c3ed2134306aab`	`90403058a9110f9d577`
`slide_img_xmp`	`7fd2da77757441f91dd37d965cd98b1f3821e68c21755a2f471a0af1ea10e806`	`46981f207d411d86fc8`
`slide_img_workflow`	`31fb6991ffb0c6aa6dfe6983909eb075de672ac427f19df31c2046f1c939556c`	`8138605b18a30fd6465`
`slide_img_timeline`	`2d71ca97b9749c255a191631f7ea62caad873b408a3202f4db1d402838ca499a`	`23b757167528383b477`
`anonymo_framework_24x24_landscape`	`5b8c4a1ba66f854ea6cc989f0bd4326094d231520be4b8561eebd30cd2e38899`	`(see final JPG in a`
`hybrid_collaboration_visual`	`945e1394264ba2dc988596c5babfa63561f47d6b0bea110c42166a2b4ef4b5d7`	`(see final JPG in a`

3.2. Distribution Packages (ZIP Archives)

The framework is distributed in versioned packages. The hashes below guarantee the integrity of the downloaded archives.

Package Name	SHA-256 Hash	Description
`anonymo_asset_package_v1.zip`	`bafc764a003fe5d05650d54a7416e0e8f6d176297f885f28f4bf3918b81093be`	Initial release: Templates and structural metadata only.
`anonymo_asset_package_v2.zip`	`69badccb3b7284fd161cb6c4c199dfd1e64dc320b2b2a848c02c7c1a0e08a8cf`	Full release: Includes the 5 high-resolution images with embedded XMP.
`I-HATR_TCF_TFB_Module_v1.0.1.zip`	`5af9b42034489830882eeaf434ee2f08b5374a9eee32bf6e13304da4cb5b2b68`	Core framework module update.

3.3. Documentation and Presentations (PDFs)

All supporting documentation generated during the hybrid workflow is also cryptographically tracked.

Document Name	SHA-256 Hash
`hybrid_provenance_en.pdf`	`facc0a50e85087fae8f2adbbc78d6cd154d6ba9dbd5b17813dca774df8cb1149`
`hybrid_provenance_pt.pdf`	`1fc36c0ae43c22eec7e7f10ff8db5de5185d04a8e3f61c2de32bf4272bef9ca4`
`hybrid_provenance_technical_en.pdf`	`f425c5afded32275bfb4f77a8bd20edce09f6adee25ab4b9280e1f8d11a5d073`
`asset_package_comparison_report.pdf`	`2425c1267811c000a3486ed9b8a493ea31fa8c6ab5bdf7601c8f92d625a36980`

4. Conclusion

By maintaining a strict separation between the visual binary hash and the metadata container, the Anonymo AI Framework provides a mathematically sound proof of existence. This ledger, combined with the public GitHub repository and the Zenodo DOI, ensures that the hybrid authorship of Chris Montgomery and Manus AI is permanently and verifiably recorded.